

BACKUP

➤ Perform Hot backup.

A Hot Backup is taken while the database is OPEN and available to users. It requires the database to be in ARCHIVELOG mode.

STEP 1: Verify ARCHIVELOG Mode.

```
SQL> archive log list;
Database log mode           Archive Mode
Automatic archival          Enabled
Archive destination         USE_DB_RECOVERY_FILE_DEST
Oldest online log sequence  20
Next log sequence to archive 22
Current log sequence        22
SQL>
SQL> select name,log_mode from v$database;
```

NAME	LOG_MODE
ORCL	ARCHIVELOG

STEP 2: Check Datafiles and Tablespaces

```
SQL>
SQL> SELECT TABLESPACE_NAME,FILE_NAME FROM DBA_DATA_FILES;
```

TABLESPACE_NAME	FILE_NAME
SYSTEM	/u01/app/oracle/oradata/ORCL/system01.dbf
SYSaux	/u01/app/oracle/oradata/ORCL/sysaux01.dbf
USERS	/u01/app/oracle/oradata/ORCL/users01.dbf
UNDOTBS1	/u01/app/oracle/oradata/ORCL/undotbs01.dbf

STEP 3: Put Tablespace in Backup Mode

Example for USERS tablespace:

```
SQL> ALTER TABLESPACE USERS BEGIN BACKUP;
Tablespace altered.
```

STEP 4: Check backup status

```
SQL> SELECT FILE#,STATUS FROM V$BACKUP;
```

FILE#	STATUS
1	NOT ACTIVE
3	NOT ACTIVE
4	NOT ACTIVE
7	ACTIVE

STEP 5: Copy the Datafile at OS Level

```
SQL> SELECT TABLESPACE_NAME,FILE_NAME FROM DBA_DATA_FILES WHERE TABLESPACE_NAME='USERS';
```

TABLESPACE_NAME	FILE_NAME
USERS	/u01/app/oracle/oradata/ORCL/users01.dbf

```
[oracle@localhost u01]$  
[oracle@localhost u01]$ cp /u01/app/oracle/oradata/ORCL/users01.dbf backup  
[oracle@localhost u01]$ ll  
total 4  
drwxrwxr-x. 4 oracle oinstall 40 Jun 6 00:22 app  
drwxrwxr-x. 2 oracle oinstall 25 Jun 10 16:07 backup  
-rw-rw-r--. 1 oracle oracle 853 Jun 9 16:13 user_creation.sql  
[oracle@localhost u01]$ cd backup  
[oracle@localhost backup]$ ll  
total 5128  
-rw-r-----. 1 oracle oracle 5251072 Jun 10 16:07 users01.dbf  
[oracle@localhost backup]$ |
```

STEP 6: Backup Control File

```
SQL> ALTER DATABASE BACKUP CONTROLFILE TO '/u01/backup/controlfile.bkp';
```

```
Database altered.
```

```
SQL> !
```

```
[oracle@localhost backup]$ ll  
total 15480  
-rw-rw----. 1 oracle oracle 10600448 Jun 10 16:17 controlfile.bkp  
-rw-r-----. 1 oracle oracle 5251072 Jun 10 16:13 users01.dbf  
[oracle@localhost backup]$ |
```

STEP 7: Create a trace file

```
SQL> ALTER DATABASE BACKUP CONTROLFILE TO TRACE;
```

```
Database altered.
```

STEP 8: Force Archive Log Switch & Backup Archived Logs

```
SQL> ALTER SYSTEM SWITCH LOGFILE;
```

```
\  
System altered.
```

```

[oracle@localhost u01]$
[oracle@localhost u01]$ cp -r /u01/app/oracle/fast_recovery_area/ORCL/archivelog/* backup
[oracle@localhost u01]$
[oracle@localhost u01]$ cd backup
[oracle@localhost backup]$ ll
total 15488
drwxr-x---. 2 oracle oracle 4096 Jun 10 16:23 2026_06_06
drwxrwx---. 2 oracle oracle 4096 Jun 10 16:23 2026_06_07
drwxrwx---. 2 oracle oracle 38 Jun 10 16:23 2026_06_08
drwxrwx---. 2 oracle oracle 38 Jun 10 16:23 2026_06_09
drwxrwx---. 2 oracle oracle 70 Jun 10 16:23 2026_06_10
-rw-rw----. 1 oracle oracle 10600448 Jun 10 16:17 controlfile.bkp
-rw-r-----. 1 oracle oracle 5251072 Jun 10 16:13 users01.dbf
[oracle@localhost backup]$ |

```

STEP 6: End Backup Mode.

```

SQL> ALTER TABLESPACE USERS END BACKUP;

Tablespace altered.

SQL> SELECT FILE#, STATUS FROM V$BACKUP;

  FILE# STATUS
-----
1 NOT ACTIVE
3 NOT ACTIVE
4 NOT ACTIVE
7 NOT ACTIVE

```

Status should show: NOT ACTIVE.

Important Interview Point

Hot Backup requires:

- Database OPEN
- ARCHIVELOG mode enabled
- Tablespaces placed in BEGIN BACKUP mode before copying datafiles
- Archived logs and control file must also be backed up
- After copying, execute END BACKUP to avoid excessive redo generation.

➤ **Perform Cold backup.**

A Cold Backup is taken when the database is completely shut down. Since no changes occur during the backup, it is the simplest and safest backup method.

Step 1: Check Database Name and File Locations

```
SELECT NAME FROM V$DATABASE;
```

```
SELECT FILE_NAME FROM DBA_DATA_FILES;
```

```
SELECT MEMBER FROM V$LOGFILE;
```

```
SHOW PARAMETER control_files;
```

STEP 2: Shutdown the Database.

STEP 3: Create Backup Directory

```
mkdir -p /u01/cold_backup
```

STEP 4: Copy Database Files, CONTROL FILES, REDO LOG FILES, PASSWORD FILES AND PARAMETER FILE.

STEP 5: Verify Backup Files

STEP 6: Start the database.

One-Line Cold Backup Procedure (Interview Answer)

1. Shutdown database using SHUTDOWN IMMEDIATE.
2. Copy all datafiles, control files, redo logs, SPFILE, and password file to backup location.
3. Start database using STARTUP.
4. Verify database is open and backup files are present.

➤ How to configure retention policy to redundancy 2?

STEP 1: check the value.

```
RMAN> show all;

using target database control file instead of recovery catalog
RMAN configuration parameters for database with db_unique_name ORCL are:
CONFIGURE RETENTION POLICY TO REDUNDANCY 1; # default
CONFIGURE BACKUP OPTIMIZATION OFF; # default
CONFIGURE DEFAULT DEVICE TYPE TO DISK; # default
CONFIGURE CONTROLFILE AUTOBACKUP ON; # default
CONFIGURE CONTROLFILE AUTOBACKUP FORMAT FOR DEVICE TYPE DISK TO '%F'; # default
CONFIGURE DEVICE TYPE DISK PARALLELISM 1 BACKUP TYPE TO BACKUPSET; # default
CONFIGURE DATAFILE BACKUP COPIES FOR DEVICE TYPE DISK TO 1; # default
CONFIGURE ARCHIVELOG BACKUP COPIES FOR DEVICE TYPE DISK TO 1; # default
CONFIGURE MAXSETSIZE TO UNLIMITED; # default
CONFIGURE ENCRYPTION FOR DATABASE OFF; # default
CONFIGURE ENCRYPTION ALGORITHM 'AES128'; # default
CONFIGURE COMPRESSION ALGORITHM 'BASIC' AS OF RELEASE 'DEFAULT' OPTIMIZE FOR LOAD TRUE ; # default
CONFIGURE RMAN OUTPUT TO KEEP FOR 7 DAYS; # default
CONFIGURE ARCHIVELOG DELETION POLICY TO NONE; # default
CONFIGURE SNAPSHOT CONTROLFILE NAME TO '/u01/app/oracle/product/19.0.0/dbhome_1/dbs/snapcf_orcl.f'; # default

RMAN> |
```

STEP 2: change the value 2

```
RMAN> CONFIGURE RETENTION POLICY TO REDUNDANCY 2;

new RMAN configuration parameters:
CONFIGURE RETENTION POLICY TO REDUNDANCY 2;
new RMAN configuration parameters are successfully stored

RMAN> show all;

RMAN configuration parameters for database with db_unique_name ORCL are:
CONFIGURE RETENTION POLICY TO REDUNDANCY 2;
CONFIGURE BACKUP OPTIMIZATION OFF; # default
CONFIGURE DEFAULT DEVICE TYPE TO DISK; # default
CONFIGURE CONTROLFILE AUTOBACKUP ON; # default
CONFIGURE CONTROLFILE AUTOBACKUP FORMAT FOR DEVICE TYPE DISK TO '%F'; # default
CONFIGURE DEVICE TYPE DISK PARALLELISM 1 BACKUP TYPE TO BACKUPSET; # default
CONFIGURE DATAFILE BACKUP COPIES FOR DEVICE TYPE DISK TO 1; # default
CONFIGURE ARCHIVELOG BACKUP COPIES FOR DEVICE TYPE DISK TO 1; # default
CONFIGURE MAXSETSIZE TO UNLIMITED; # default
CONFIGURE ENCRYPTION FOR DATABASE OFF; # default
CONFIGURE ENCRYPTION ALGORITHM 'AES128'; # default
CONFIGURE COMPRESSION ALGORITHM 'BASIC' AS OF RELEASE 'DEFAULT' OPTIMIZE FOR LOAD TRUE ; # default
CONFIGURE RMAN OUTPUT TO KEEP FOR 7 DAYS; # default
CONFIGURE ARCHIVELOG DELETION POLICY TO NONE; # default
CONFIGURE SNAPSHOT CONTROLFILE NAME TO '/u01/app/oracle/product/19.0.0/dbhome_1/dbs/snapcf_orcl.f'; # default

RMAN> |
```

➤ How to configure control file auto backup ON?

```
RMAN> show all;

RMAN configuration parameters for database with db_unique_name ORCL are:
CONFIGURE RETENTION POLICY TO REDUNDANCY 2;
CONFIGURE BACKUP OPTIMIZATION OFF; # default
CONFIGURE DEFAULT DEVICE TYPE TO DISK; # default
CONFIGURE CONTROLFILE AUTOBACKUP ON; # default
CONFIGURE CONTROLFILE AUTOBACKUP FORMAT FOR DEVICE TYPE DISK TO '%F'; # default
CONFIGURE DEVICE TYPE DISK PARALLELISM 1 BACKUP TYPE TO BACKUPSET; # default
CONFIGURE DATAFILE BACKUP COPIES FOR DEVICE TYPE DISK TO 1; # default
CONFIGURE ARCHIVELOG BACKUP COPIES FOR DEVICE TYPE DISK TO 1; # default
CONFIGURE MAXSETSIZE TO UNLIMITED; # default
CONFIGURE ENCRYPTION FOR DATABASE OFF; # default
CONFIGURE ENCRYPTION ALGORITHM 'AES128'; # default
CONFIGURE COMPRESSION ALGORITHM 'BASIC' AS OF RELEASE 'DEFAULT' OPTIMIZE FOR LOAD TRUE ; # default
CONFIGURE RMAN OUTPUT TO KEEP FOR 7 DAYS; # default
CONFIGURE ARCHIVELOG DELETION POLICY TO NONE; # default
CONFIGURE SNAPSHOT CONTROLFILE NAME TO '/u01/app/oracle/product/19.0.0/dbhome_1/dbs/snapcf_orcl.f'; # default
```

➤ How to delete obsolete and Expired backup via RMAN?

OBSOLETE BACKUP

STEP 1: Check the obsolete backup.

```
RMAN> REPORT OBSOLETE;  
  
RMAN retention policy will be applied to the command  
RMAN retention policy is set to redundancy 2  
no obsolete backups found
```

STEP 2: Delete obsolete backup.

```
RMAN> DELETE OBSOLETE;  
  
RMAN retention policy will be applied to the command  
RMAN retention policy is set to redundancy 2  
allocated channel: ORA_DISK_1  
channel ORA_DISK_1: SID=15 device type=DISK  
no obsolete backups found
```

EXPIRED BACKUP

STEP 1: Check expired backup.

```
RMAN> CROSSCHECK BACKUP;  
  
using channel ORA_DISK_1  
crosschecked backup piece: found to be 'AVAILABLE'  
backup piece handle=/u01/app/oracle/fast_recovery_area/ORCL/autobackup/2026_06_06/o1_mf_s_1235225284_o27q3g46_.bkp RECID=1 STA  
MP=1235225286  
crosschecked backup piece: found to be 'AVAILABLE'  
backup piece handle=/u01/app/oracle/fast_recovery_area/ORCL/backupset/2026_06_06/o1_mf_anndf_TAG20260606T144322_o27s5n84_.bkp  
RECID=2 STAMP=1235227403  
crosschecked backup piece: found to be 'AVAILABLE'  
backup piece handle=/u01/app/oracle/fast_recovery_area/ORCL/backupset/2026_06_06/o1_mf_nnndf_TAG20260606T144327_o27s5qlx_.bkp  
RECID=3 STAMP=1235227407  
crosschecked backup piece: found to be 'AVAILABLE'  
backup piece handle=/u01/app/oracle/fast_recovery_area/ORCL/backupset/2026_06_06/o1_mf_anndf_TAG20260606T144404_o27s6wr3_.bkp  
RECID=4 STAMP=1235227444  
crosschecked backup piece: found to be 'AVAILABLE'  
backup piece handle=/u01/app/oracle/fast_recovery_area/ORCL/autobackup/2026_06_06/o1_mf_s_1235227447_o27s6zws_.bkp RECID=5 STA  
MP=1235227447  
Crosschecked 5 objects  
  
RMAN> |
```

STEP 2: Delete expired backup.

RMAN> DELETE EXPIRED BACKUP;

➤ How to take Incremental 1 level (Cumulative & Differential) & 0 level incremental backup?

STEP 1: Level 0 Incremental Backup

A **Level 0 backup** is the base backup. It copies all used blocks in the database.

```
RMAN>      BACKUP      INCREMENTAL      LEVEL      0      DATABASE      FORMAT  
'/u02/backup/L0_%d_%T_%U.bkp';
```

```
RMAN> BACKUP INCREMENTAL LEVEL 0 DATABASE FORMAT '/u02/backup/L0_%d_%T_%U.bkp';  
  
Starting backup at 11-JUN-26  
using channel ORA_DISK_1  
channel ORA_DISK_1: starting incremental level 0 datafile backup set  
channel ORA_DISK_1: specifying datafile(s) in backup set  
input datafile file number=000001 name=/u01/app/oracle/oradata/ORCL/system01.dbf  
input datafile file number=000003 name=/u01/app/oracle/oradata/ORCL/sysaux01.dbf  
input datafile file number=000004 name=/u01/app/oracle/oradata/ORCL/undotbs01.dbf  
input datafile file number=000007 name=/u01/app/oracle/oradata/ORCL/users01.dbf  
channel ORA_DISK_1: starting piece 1 at 11-JUN-26  
channel ORA_DISK_1: finished piece 1 at 11-JUN-26  
piece handle=/u02/backup/L0_ORCL_20260611_074qdrdv_1_1.bkp tag=TAG20260611T193007 comment=NONE  
channel ORA_DISK_1: backup set complete, elapsed time: 00:00:37  
Finished backup at 11-JUN-26  
  
Starting Control File and SPFILE Autobackup at 11-JUN-26  
piece handle=/u01/app/oracle/fast_recovery_area/ORCL/autobackup/2026_06_11/o1_mf_s_1235676645_o2ohwg1r_.bkp comment=NONE  
Finished Control File and SPFILE Autobackup at 11-JUN-26
```

STEP 2: Level 1 Differential Incremental Backup

A **Level 1 Differential** backs up blocks changed since the most recent Level 0 or Level 1 backup.

Example Schedule

Day	Backup
Sunday	Level 0
Monday	Level 1 Differential
Tuesday	Level 1 Differential
Wednesday	Level 1 Differential

```
RMAN>BACKUP INCREMENTAL LEVEL 1 DATABASE FORMAT
'/u02/backup/L1_%d_%T_%U.bkp';
```

```
RMAN> BACKUP INCREMENTAL LEVEL 1 DATABASE FORMAT '/u02/backup/L1_%d_%T_%U.bkp';

Starting backup at 11-JUN-26
using channel ORA_DISK_1
channel ORA_DISK_1: starting incremental level 1 datafile backup set
channel ORA_DISK_1: specifying datafile(s) in backup set
input datafile file number=00001 name=/u01/app/oracle/oradata/ORCL/system01.dbf
input datafile file number=00003 name=/u01/app/oracle/oradata/ORCL/sysaux01.dbf
input datafile file number=00004 name=/u01/app/oracle/oradata/ORCL/undotbs01.dbf
input datafile file number=00007 name=/u01/app/oracle/oradata/ORCL/users01.dbf
channel ORA_DISK_1: starting piece 1 at 11-JUN-26
channel ORA_DISK_1: finished piece 1 at 11-JUN-26
piece handle=/u02/backup/L1_ORCL_20260611_094qdrj1_1_1.bkp tag=TAG20260611T193248 comment=NONE
channel ORA_DISK_1: backup set complete, elapsed time: 00:00:07
Finished backup at 11-JUN-26

Starting Control File and SPFILE Autobackup at 11-JUN-26
piece handle=/u01/app/oracle/fast_recovery_area/ORCL/autobackup/2026_06_11/o1_mf_s_1235676776_o2oj0jvr_.bkp comment=NONE
Finished Control File and SPFILE Autobackup at 11-JUN-26
```

STEP 3: Level 1 Cumulative Incremental Backup.

A Level 1 Cumulative backs up all blocks changed since the last Level 0 backup.

Example Schedule

Day	Backup
Sunday	Level 0
Monday	Level 1 Cumulative
Tuesday	Level 1 Cumulative
Wednesday	Level 1 Cumulative

```
RMAN> BACKUP INCREMENTAL LEVEL 1 CUMULATIVE DATABASE FORMAT
'/u02/backup/LC1__%d_%T_%U.bkp';
```

```
RMAN> BACKUP INCREMENTAL LEVEL 1 CUMULATIVE DATABASE FORMAT '/u02/backup/LC1__%d_%T_%U.bkp';

Starting backup at 11-JUN-26
using channel ORA_DISK_1
channel ORA_DISK_1: starting incremental level 1 datafile backup set
channel ORA_DISK_1: specifying datafile(s) in backup set
input datafile file number=00001 name=/u01/app/oracle/oradata/ORCL/system01.dbf
input datafile file number=00003 name=/u01/app/oracle/oradata/ORCL/sysaux01.dbf
input datafile file number=00004 name=/u01/app/oracle/oradata/ORCL/undotbs01.dbf
input datafile file number=00007 name=/u01/app/oracle/oradata/ORCL/users01.dbf
channel ORA_DISK_1: starting piece 1 at 11-JUN-26
channel ORA_DISK_1: finished piece 1 at 11-JUN-26
piece handle=/u02/backup/LC1_ORCL_20260611_0b4qdrva_1_1.bkp tag=TAG20260611T193922 comment=NONE
channel ORA_DISK_1: backup set complete, elapsed time: 00:00:07
Finished backup at 11-JUN-26

Starting Control File and SPFILE Autobackup at 11-JUN-26
piece handle=/u01/app/oracle/fast_recovery_area/ORCL/autobackup/2026_06_11/o1_mf_s_1235677169_o2ojdsv4_.bkp comment=NONE
Finished Control File and SPFILE Autobackup at 11-JUN-26
```


Recovery Requirement

To recover Wednesday:

- Level 0
- Wednesday Cumulative

Only two backups are required.

✓ Differential backup stores changes since the last incremental backup, whereas cumulative backup stores all changes since the last Level 0 backup. Differential backups are smaller and faster; cumulative backups simplify recovery.

➤ How to Configure RMAN Default Channels?

A channel is a connection between RMAN and the database that RMAN uses to perform backup and recovery operations.

STEP 1: Clear the existing channel.

```
RMAN> CONFIGURE CHANNEL DEVICE TYPE DISK CLEAR;  
old RMAN configuration parameters are successfully deleted
```

STEP 2: Configure the new channel.

```
RMAN> CONFIGURE CHANNEL DEVICE TYPE DISK FORMAT '/u02/backup/%U.bkp';  
new RMAN configuration parameters:  
CONFIGURE CHANNEL DEVICE TYPE DISK FORMAT    '/u02/backup/%U.bkp';  
new RMAN configuration parameters are successfully stored
```

STEP 3: Check the channel.

```
RMAN> show channel;  
RMAN configuration parameters for database with db_unique_name ORCL are:  
CONFIGURE CHANNEL DEVICE TYPE DISK FORMAT    '/u02/backup/%U.bkp';
```

➤ **How to take Compressed RMAN Backups in RMAN?**

STEP 1: Compressed Full Database Backup.

RMAN> backup as compressed backupset database;

STEP 2: Compressed Database + Archive log Backup.

RMAN> backup as compressed backupset database PLUS ARCHIVELOG;

STEP 3: Compressed Level 0 Incremental Backup.

RMAN> backup as compressed backupset incremental level 0 database;

STEP 4: Compressed Level 1 Differential Backup

RMAN> backup as compressed backupset incremental level 1 database;

STEP 5: Compressed Level 1 Cumulative Backup

RMAN> backup as compressed backupset incremental level 1 cumulative database;

➤ **Command to check backup status, its completion & running status?**

```
SQL> SELECT SESSION_KEY,INPUT_TYPE,STATUS,START_TIME,END_TIME FROM V$RMAN_BACKUP_JOB_DETAILS ORDER BY START_TIME DESC;
```

SESSION_KEY	INPUT_TYPE	STATUS	START_TIM	END_TIME
12	DB INCR	FAILED	11-JUN-26	11-JUN-26
1	DB FULL	COMPLETED	06-JUN-26	06-JUN-26

STATUS values:

- COMPLETED
- COMPLETED WITH WARNINGS
- FAILED
- RUNNING

➤ **How to delete archive log keeping 2 days logs?**

STEP 1: Delete Archive logs Older Than 2 Days

RMAN> DELETE ARCHIVELOG ALL COMPLETED BEFORE 'SYSDATE-2';

STEP 2: Delete Archive logs Older Than 2 Days (WITHOUT CONFIRMATION)

```
RMAN> DELETE NOPROMPT ARCHIVELOG ALL COMPLETED BEFORE 'SYSDATE-2';
```

➤ **How to kill/disconnect RMAN Backup session?**

STEP 1: Check the RMAN session

```
SELECT SID,  
       SERIAL#,  
       PROGRAM,  
       STATUS  
FROM V$SESSION  
WHERE PROGRAM LIKE '%rman%';
```

STEP 2: Kill the RMAN Session

```
ALTER SYSTEM KILL SESSION '123,45678' IMMEDIATE;
```

➤ **How to backup control file to trace?**

STEP 1: To back up the control file to a trace file, connect as SYSDBA and run

```
RMAN> ALTER DATABASE BACKUP CONTROLFILE TO TRACE;  
  
Statement processed
```

STEP 2: Verify the Trace File Location.

```
RMAN> SELECT VALUE FROM V$DIAG_INFO WHERE NAME='Diag Trace';
```

```
VALUE  
-----
```

```
/u01/app/oracle/diag/rdbms/orcl/orcl/trace
```

STEP 3: Go to the trace directory.

```
cd /u01/app/oracle/diag/rdbms/orcl/ORCL/trace
```

```
ls -ltr *.trc
```

The latest .trc file will contain the control file recreation script.